Bangladesh University of Engineering and Technology

Course no: ME-310
Course Title: Thermo-Fluid System Design



Portable Cooling Pad: A refrigeration Cycle for Laptop

Team Members



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Problem Statement

Laptop overheating is a common issue faced by many users, leading to reduced performance, potential hardware damage, and discomfort during use.



As Mechanical Engineering students, we often have to work with heavy software such as CAD applications or simulations tools, which demand high computational power. As a result, overheating of laptop become a inseparable part of our lives.

Available Solutions and Their Limitations



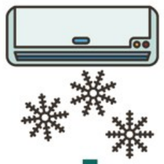
Integratred Cooling System

It is a built in technology in laptop, can not cool down laptops when temperature rises drastically high while gaming or using heavy softwares.



External Cooling Fan

Uses Forced convection to cool down laptops, But it carries dust to laptops and eventually blocks the ventilation system.

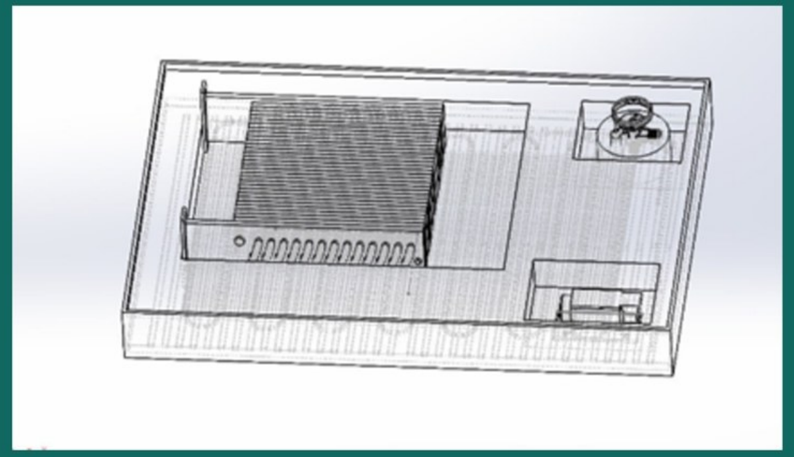


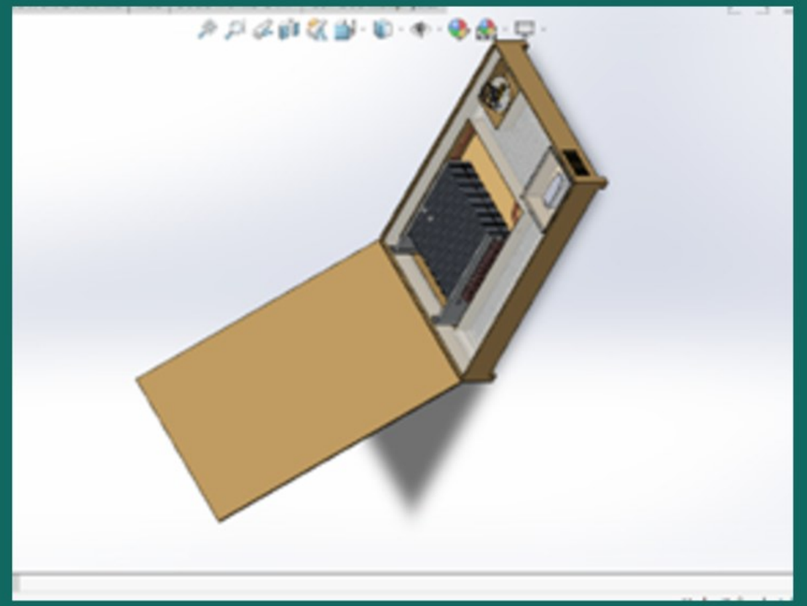
Room Cooling Solutions

A passive cooling system that use an air conditioner to lower the ambient temperature. But it is too costly.

Our Solution

Our proposed solution is using the refrigeration cycle to cool down laptop. Refrigeration cycle is more efficient than fan type cooler. It can cool down the temperature of laptop about 10-15°C. We will use a portable cooling pad, having a dimension of 14″ x10″ x3″ which incorporate the refrigeration cycle. The length of the condenser and evaporator coil will be 2.78 m and 5.56 m respectively.



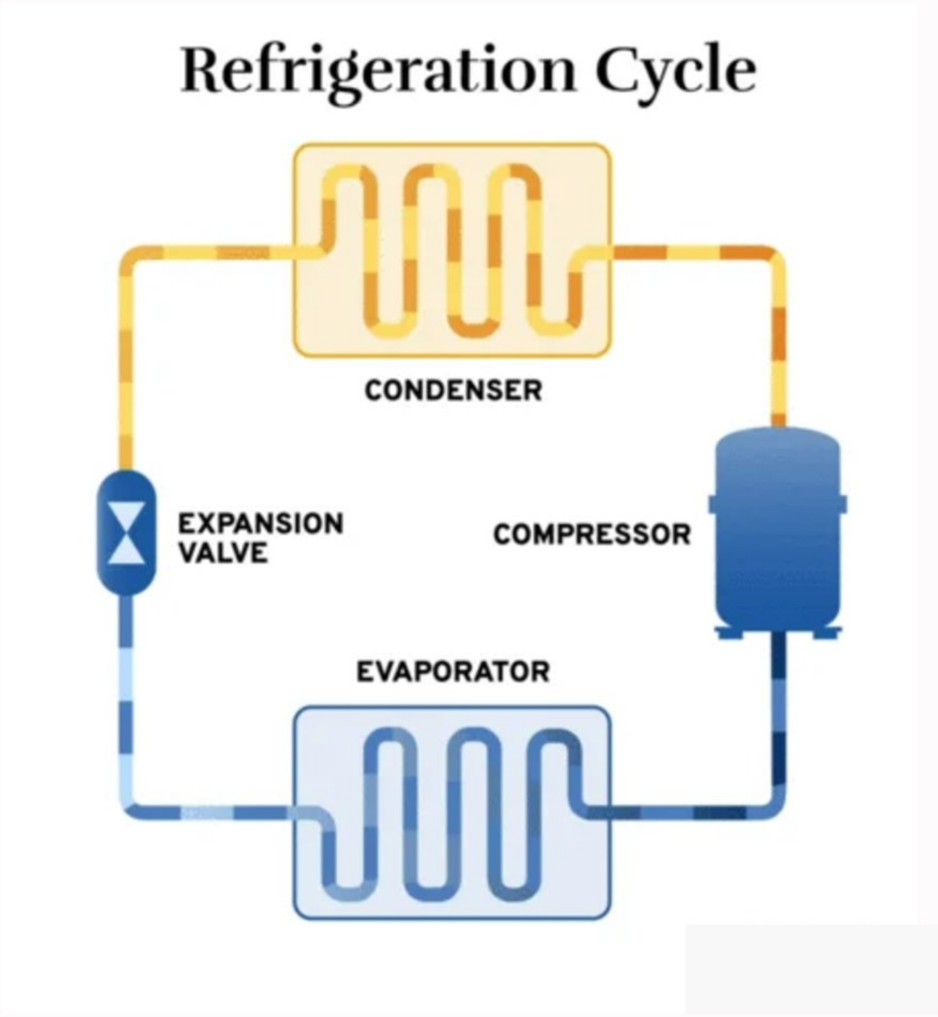


Introduction to Refigeration Cycle

The refrigeration cycle is a fundamental process used in refrigeration and air conditioning systems to transfer heat from one area to another, **thereby cooling a space or substance. The cycle typically operates using a refrigerant as a working fluid and follows these four main stages:**

Key Components

1. **Compressor**: Pressurizes the refrigerant.
2. **Condenser**: Rejects heat and converts the refrigerant vapor into liquid.
3. **Expansion Valve**: Reduces the refrigerant's pressure and temperature.
4. **Evaporator:** Absorbs heat from the space or substance being cooled.



Objectives

01



Design and Development

02



Performance Enhancement

03



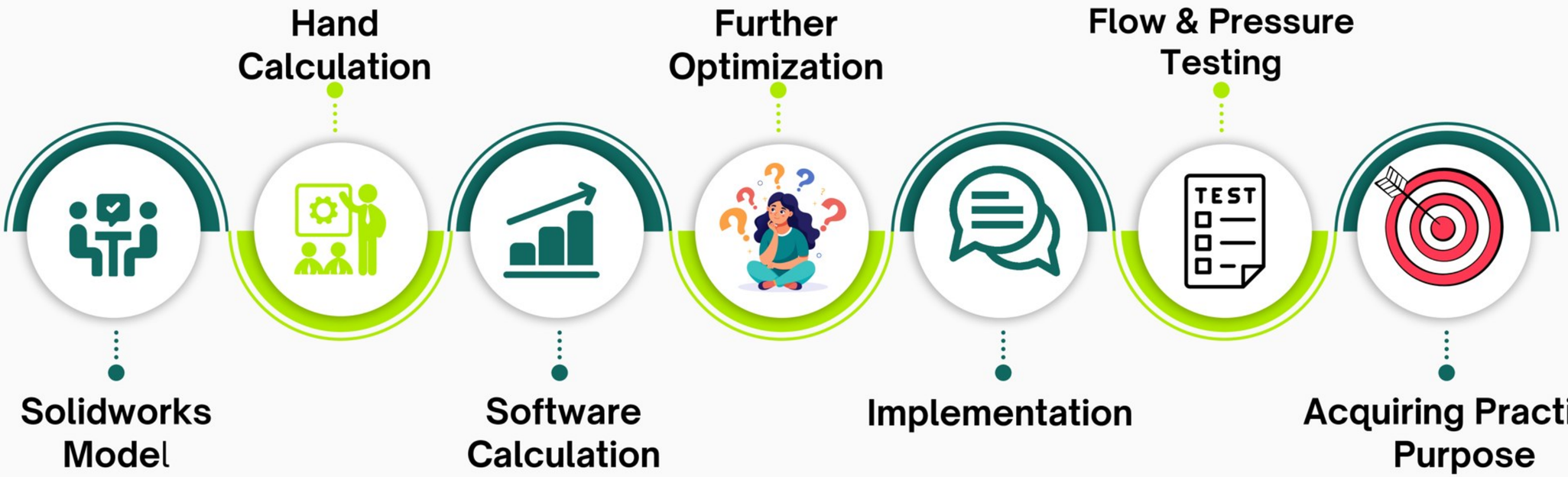
Energy Efficiency

04



Portability and Usability

Design Strategy



Design Parameters

Evaporator:

Length: 5.56m
Diameter: 9.5m
Thickness: 1 mm
Material: Cu

Condenser:

Length: 2.78m
Diameter: 9.5 mm
Coolant: R-134A
Mass flow rate: 2.93 m/s

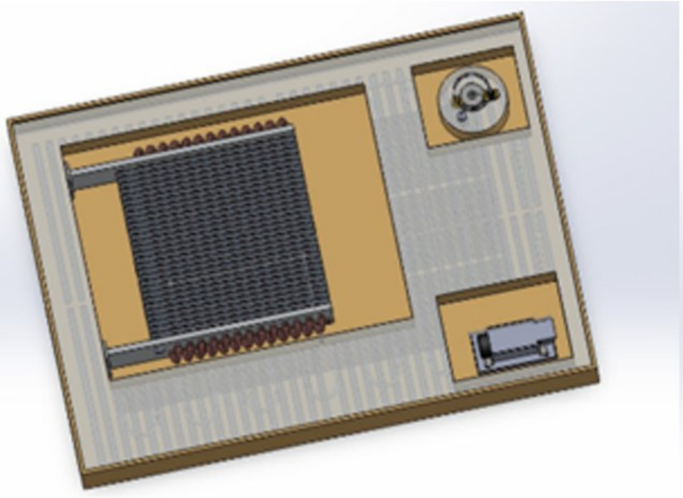
Expansion Valve:

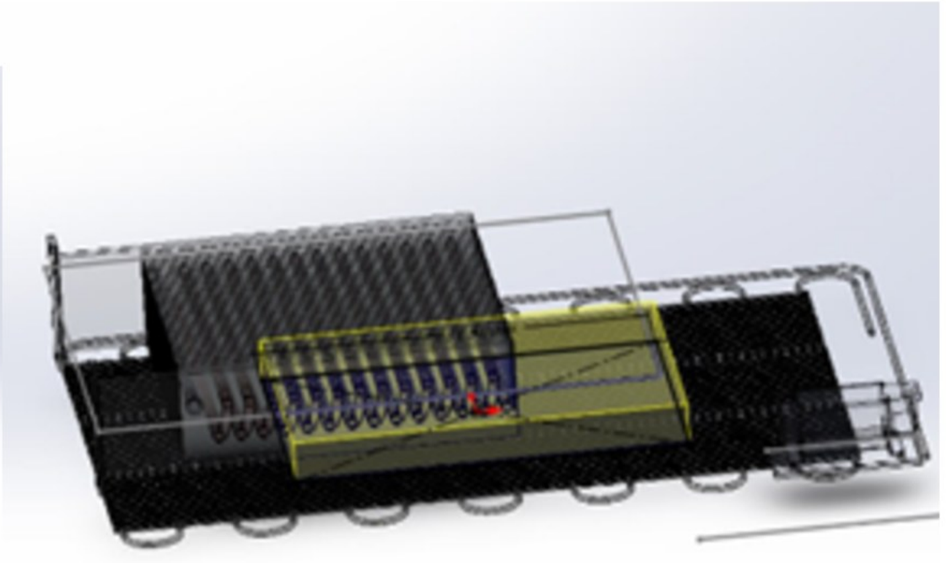
Flow capacity:
10-12 kg/h
Orifice size: 0.5-1 mm
Refrigerant: R-134A

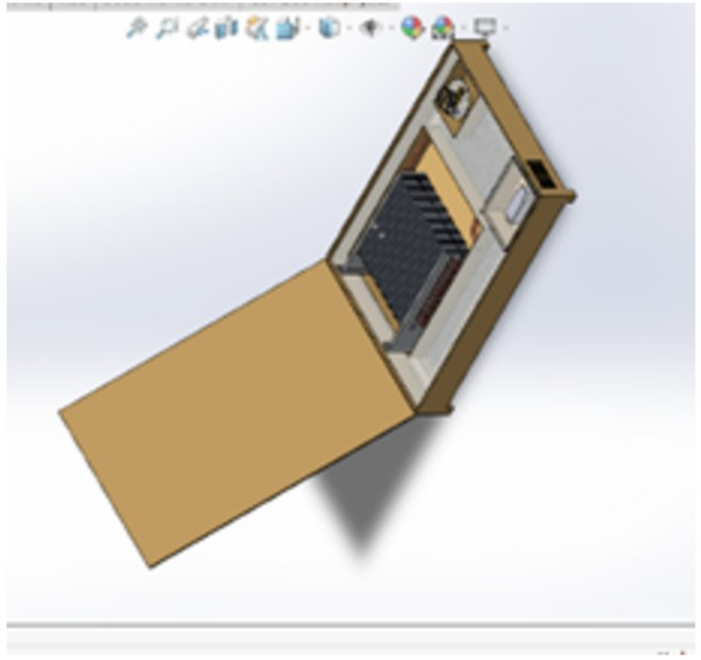
Compressor:

Power=50W

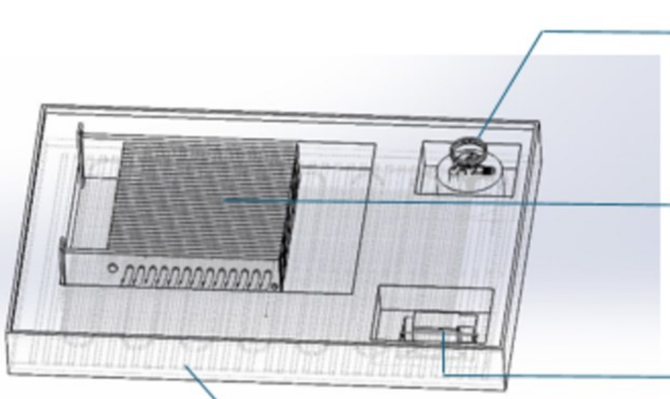
CAD Model

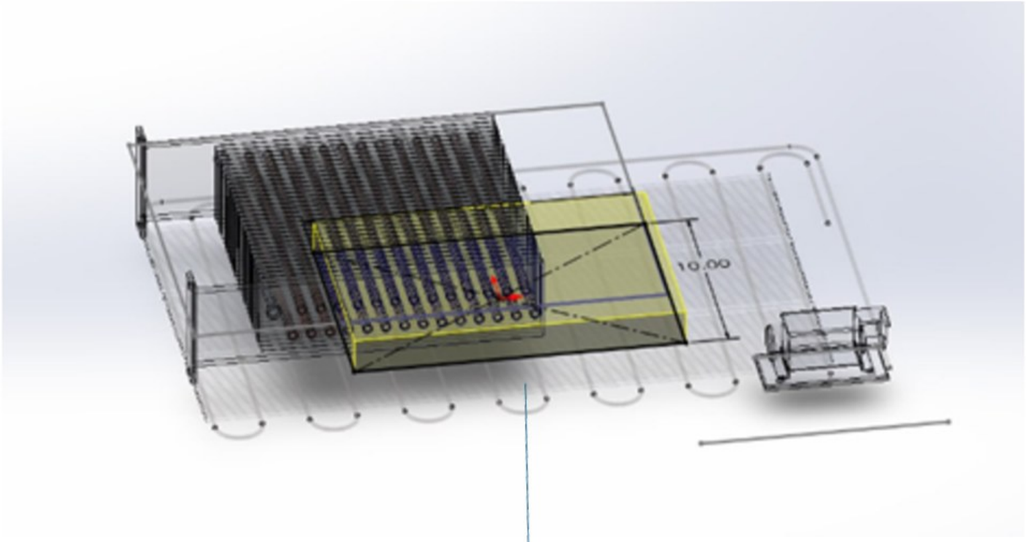






CAD Model







Condenser

fig: Different parts of the cooler

Mathematical Modeling



In this project, the evaporator and condenser tube consist of cylindrical thin tubes in which the refrigerant fluid passes that is mechanically supported by a mounting frame.

To mathematically model such a complex device, we assume that the flow rates of the fluids are constant. We also accept that the inlet temperatures of the fluid are unchanging. Moreover, it is taken for granted that the geometric structure and property of both condenser and evaporator tube material are homogenous.

Software Calculation

```python
import math

# Constants for R-134A Refrigerant
R134A_PROPERTIES = {
    "thermal_conductivity": 0.08,  # W/m-K
    "specific_heat": 1420,         # J/kg-K
    "density": 1200,               # kg/m^3 (approx. liquid phase)
    "latent_heat": 200000,         # J/kg
}

# Constants for Copper Tube
CU_PROPERTIES = {
    "thermal_conductivity": 401,  # W/m-K
}

# System Parameters
mass_flow_rate = 0.02  # kg/s (example)
refrigerant_inlet_temp = 40  # °C (for condenser)
refrigerant_exit_temp = 10   # °C (for evaporator)
ambient_temp = 25  # °C (for condenser surroundings)

# Heat Transfer Coefficient Calculation
def heat_transfer_coefficient(k, d):
    """Calculates the heat transfer coefficient."""
    return k / d

# Function to Calculate Pipe Length for Condenser
def calculate_condenser_length(diameter, heat_transfer_rate):
    """Calculates the required pipe length for the condenser."""
    h = heat_transfer_coefficient(R134A_PROPERTIES["thermal_conductivity"], diameter)
    delta_t = refrigerant_inlet_temp - ambient_temp
    length = heat_transfer_rate / (math.pi * diameter * h * delta_t)
    return length

# Function to Calculate Pipe Length for Evaporator
def calculate_evaporator_length(diameter, heat_transfer_rate):
    """Calculates the required pipe length for the evaporator."""
    h = heat_transfer_coefficient(R134A_PROPERTIES["thermal_conductivity"], diameter)
    delta_t = refrigerant_exit_temp - ambient_temp
    length = heat_transfer_rate / (math.pi * diameter * h * delta_t)
    return length
```

```python
# Iterative Optimization for Dimensions
def optimize_dimensions(heat_transfer_rate, target_length, component):
    """Finds the optimal pipe dimensions for a given component."""
    diameter = 0.005  # Initial guess for diameter (m)
    for _ in range(100):  # Limit iterations
        if component == "condenser":
            length = calculate_condenser_length(diameter, heat_transfer_rate)
        elif component == "evaporator":
            length = calculate_evaporator_length(diameter, heat_transfer_rate)
        else:
            raise ValueError("Invalid component type")

        if abs(length - target_length) < 0.01:  # Target length tolerance
            break
        diameter += 0.0001  # Increment diameter (0.1 mm)

    return diameter, length

# Heat Transfer Rate Calculation
heat_transfer_rate_condenser = mass_flow_rate * R134A_PROPERTIES["latent_heat"]  # W
heat_transfer_rate_evaporator = mass_flow_rate * R134A_PROPERTIES["latent_heat"]  # W

# Target Lengths
target_length_condenser = 2.78  # meters
target_length_evaporator = 5.56  # meters

# Optimize Dimensions
condenser_diameter, condenser_length = optimize_dimensions(
    heat_transfer_rate_condenser, target_length_condenser, "condenser")
evaporator_diameter, evaporator_length = optimize_dimensions(
    heat_transfer_rate_evaporator, target_length_evaporator, "evaporator")

# Results
print("Condenser Dimensions:")
print(f"  Diameter: {condenser_diameter:.4f} m")
print(f"  Length: {condenser_length:.2f} m")

print("\nEvaporator Dimensions:")
print(f"  Diameter: {evaporator_diameter:.4f} m")
print(f"  Length: {evaporator_length:.2f} m")
```

Fabrication Method

Choosing material and Component

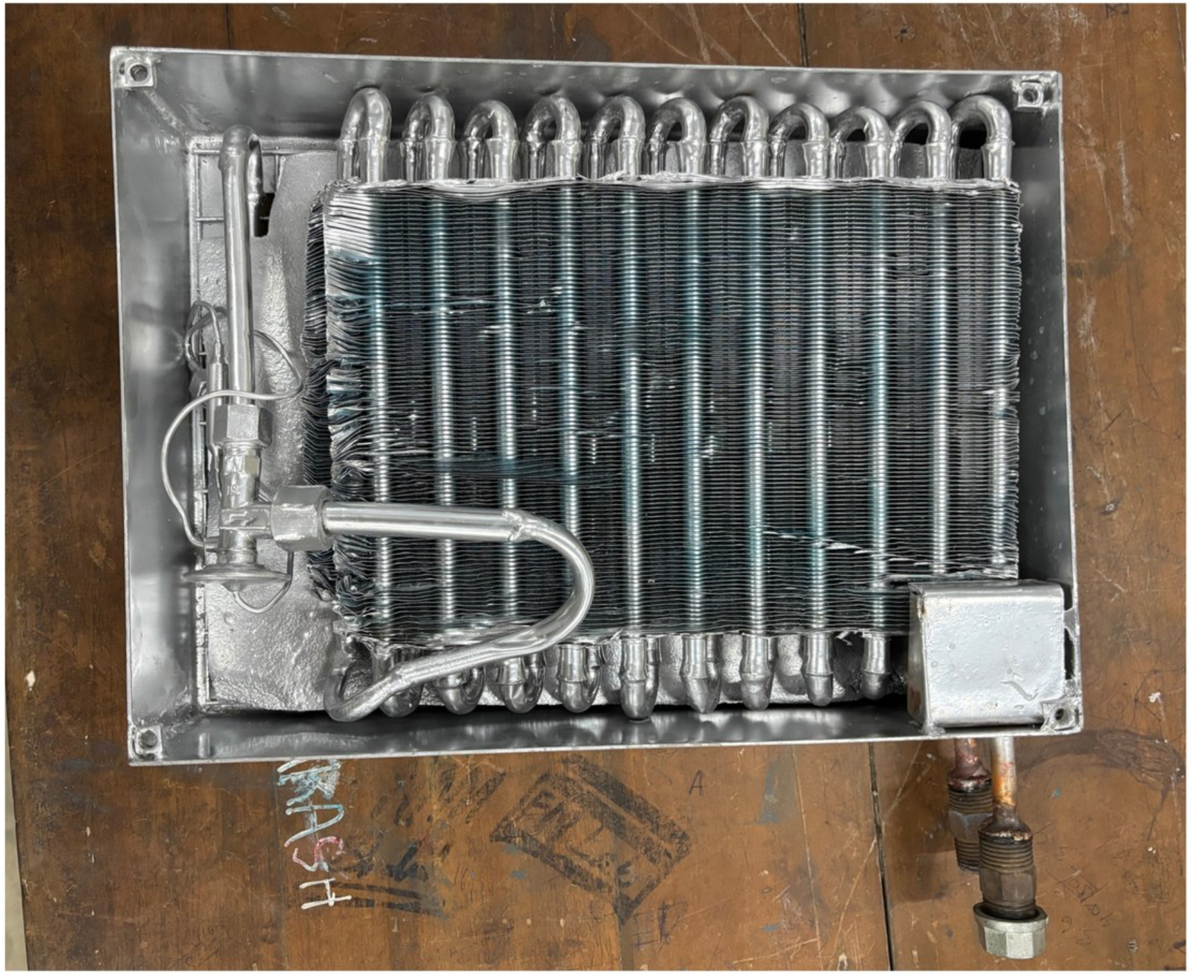
Fabrication of Condenser and Evaporator

Installation of Compressor and Expansion Valve

Welding and Assembly of all the Component in Outer Shell

Testing and Finalizing the Model

Fabricated Portable Cooling Pad for Laptop



Flow Circuit Test

After making the prototype, it was taken to **BUET Fluid Lab** to Flow circuit test. The project successfully **passed** the **flow circuit test**



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Submitted by-

Group 02, Section: A1

2010004 – Juthi Hira

2010011 – Amet Hasan Emon

2010015 – Fayez Uddin Ahmed Farhan

2010017 – Avijit Mondal Shantonu

2010029 – Kishar Das Gupta

This is to certify that **'Portable Cooling Pad: A refrigeration Cycle for Laptop'** has successfully passed the Flow Test. No leakage was observed.

Received by Fluid Lab.

Date: 13.12.24

Time: 6:30 PM Sign.....

Signature

Pressure Test

After making the prototype, it was taken to **BUET Hydraulics Lab** for a pressure test. It has successfully **passed** the test sustaining **200 psi** Hydraulic pressure.



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This is to certify that '**Portable Cooling Pad: A refrigeration Cycle for Laptop**' has successfully passed the Hydraulic Test. No leakage was observed.

(200 Psi sustained)

17/12/24

Lab Instructor, ME, Buet

Signature

Rating of the Fabricated Model

Heat Duty	150W
COP	3
Length per pass of the evaporator	9 inch

Cost Analysis

No.	Name Sector	Unit Cost (BDT)	Total Cost(BDT)
1	Evaporator	30(per ft)	6480
2	Condenser Tube	24(per ft)	2600
3	Expansion Valve		650
4	Box		400
5	Fittings		320
6	Labor		1500
7	Transports		400
8	Others		150
Total			12,500

Limitations & Possible Solutions:

1. **Complexity of Design**

Possible Solution: The system can be simplified by utilising compact pre-assembled refrigeration modules or miniature sub-assembly components.

2. **Size and Weight**

Possible Solution: Downscale the system with small micro-compressors, and small heat exchangers. and choosing alloys or composites materials.

3.**Unavailability of 50W Compressor**

Possible Solution: The compressor can be custom made or DC compressor can be used instead of 50W compressor.

Future Aspect:

- Further miniaturization of the cooling system.
- Exploration of more eco-friendly refrigerants with lower global warming potential.
- Enhancement of energy efficiency to broaden applicability.

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- **Dr. Zahurul Haque**
  Professor, Department of Mechanical Engineering, BUET
- **Dr. Mohammad Arif Hasan Mamun**
  Professor, Department of Mechanical Engineering, BUET

THANK YOU

For watching this presentation

Any Question

